Freight-rate prediction

Muftasim Fuad | Technical approach and validation

CatBoost predicts total rates for 12,000 November-December 2025 loads. The selected main model achieves **$121.83 September-October holdout MAE**, a **50.5% reduction** versus the distance x quote baseline. Systematic underprediction and uncertain future-city accuracy are the main limitations. The hidden final score is unknown.

CatBoost suits mixed numerical and categorical tabular data and learns nonlinear effects and interactions among distance, equipment, location and market inputs. It avoids manually one-hot encoding all lanes and supports numerical missing values. These conveniences do not by themselves establish performance on unseen cities.

1. Data review

Input	Coverage and role
Labeled development	48,000 rows; January 1-October 31, 2025; target: posted_rate.
Final prediction set	12,000 rows; November 1-December 31, 2025; targets withheld.
December scenario	31 daily rows; Lexington to Fort Wayne, 360 miles, Dry Van, 32,000 lb.

Training has 300 missing weights, 292 non-positive weights, and 374 missing market indices. The final set has 165 missing weights, 145 non-positive weights and 249 missing market indices. Non-positive weights become missing; separate invalidity and missingness flags are retained. CatBoost handles numerical NaNs directly.

There are no duplicate IDs or duplicate complete feature rows within either dataset. IDs do not overlap across the labeled and final sets. Targets and distances are finite and positive. Large positive target values are retained, including in evaluation: there is insufficient evidence to label them erroneous. Their impact is visible in the gap between MAE and RMSE.

Eight cities are unseen in training, affecting 1,447 final rows (12.1%). The September-October holdout has zero unseen-city loads relative to January-August training. It does not measure this part of deployment. Accepting unseen categories at inference does not demonstrate accuracy.

The root README is the solution entry point. Original challenge instructions are retained in provided_readme.md. Commands use the actual supplied root-level, hyphenated filenames.

2. Validation and model choice

Stage	Training window	Evaluation window
Candidate selection	Jan-Jun (28,806)	Jul-Aug (9,671)
Local holdout	Jan-Aug (38,477)	Sep-Oct (9,523)
Final fit	Jan-Oct (48,000)	Nov-Dec: labels unavailable

A chronological split matches the future-only prediction set. Random splitting would mix past and future market regimes. Candidate selection uses July-August MAE; September-October is kept outside model selection. After local evaluation, refit the selected models on all labeled rows. The final-set features are audited but never used for fitting, tuning or preprocessing statistics.

MAE is the declared primary criterion because the supplied scorer does not specify an accuracy metric. RMSE, WAPE and R-squared are reported as complementary diagnostics. All candidates use 650 trees, depth 6, learning rate 0.055, seed 42 and four CPU threads; no holdout early stopping.

July-August candidate	MAE ($)	RMSE ($)
Distance x quote baseline	367.29	753.96
Median baseline	1,128.11	1,532.19
All inputs / MAE	156.66	631.55
Quote residual / MAE	242.73	660.54
All inputs / RMSE	193.00	649.12
Without quote / MAE	140.48	623.99
Six inputs / total rate	163.01	628.75
Six inputs / rate per mile	159.47	627.62

Selected full-data model: **Without quote / MAE**. Selected December model: **Six inputs / rate per mile**. The quote-free candidate is eligible for the main prediction task. Its better selection MAE supports omitting quote_signal; no assumption about its upstream construction is needed by the selected model. The comparison does not prove that quote_signal is leakage.

3. Held-out results

September-October	MAE ($)	RMSE ($)	WAPE
Median baseline	1,148.92	1,569.42	48.02%
Distance x quote	246.02	711.80	10.28%
Selected main model	121.83	638.85	5.09%
Selected December model	106.34	632.95	4.44%

Main-model holdout R-squared: 0.8247; December-model R-squared: 0.8280. These are local retrospective results, not the hidden November-December score. The supplied score.py validates file format and generates the chart; only Spotter can score final accuracy.

Main-model slice	Rows	MAE ($)	RMSE ($)
2025-09	4,670	115.63	619.78
2025-10	4,853	127.79	656.69
Dry Van	5,360	116.47	659.73
Flatbed	1,770	125.84	515.13
Reefer	2,393	130.89	673.06

Feature design

Categorical pickup, delivery, equipment and directional lane; distance and log-distance; cleaned weight with flags; day of week, day of month, month, elapsed day, weekend/month-end flags and annual sine/cosine terms. The main model additionally uses supplied coordinates, coordinate differences and market index with its missingness flag. Neither IDs nor target rates are input features. All feature transformations are stateless; categorical statistics are learned inside each fit.

Practical limits

One year of January-October data cannot establish December holiday seasonality. Tree models have limited extrapolation beyond observed time trends; cyclical features do not solve missing seasonal coverage. The core model lacks contemporaneous market information. New-city generalization and future market shifts remain risks. No confidence interval or minimum final score is claimed. More history, verified feature provenance and interval calibration would be useful next steps. Additional development-only chronological and city-held-out experiments are reported in Section 8.

4. Residuals: tail risk and underprediction

The following post-evaluation diagnostics use the saved holdout predictions, joined to source rows by unique load_id. Signed error is prediction minus actual. This analysis does not fit, select or calibrate models; all predictions remain unchanged.

Main-model diagnostic	Result
MAE	$121.83
Median absolute error	$42.49
95th-percentile absolute error	$238.01
Mean signed error	-$90.93
Median signed error	-$37.83
Loads underpredicted	80.6%
Squared error from worst 96 rows (about 1%)	96.5%

Slice	Rows	Mean error ($)	Median error ($)	Underpredicted
2025-09	4,670	-88.02	-36.74	81.0%
2025-10	4,853	-93.72	-38.76	80.2%
<500 mi	2,089	-26.80	-11.00	67.0%
500-<1,000 mi	2,973	-51.88	-32.00	81.5%
1,000-<2,000 mi	2,927	-130.17	-66.30	85.2%
>=2,000 mi	1,534	-179.05	-119.65	88.8%

Two distinct issues are visible: rare extreme misses dominate squared error, while the negative median and 80.6% underprediction rate show broad underprediction among ordinary loads. Bias appears in both months and all fixed distance bands; its dollar magnitude and frequency increase with distance. These descriptive relationships do not establish the cause.

Operational consequence: using these estimates directly for quotes or budgets could understate transport costs, especially on longer trips. Extreme observations remain in all headline metrics. No $90.93 offset or other correction is estimated from this holdout. Any correction must be fit and validated within development-only chronological folds, then assessed on a separate future evaluation period.

5. Model decisions and validation gaps

Model	Jul-Aug MAE ($)	Sep-Oct MAE ($)
Main: no quote / total dollars	140.48	121.83
Core: six inputs / rate per mile	159.47	106.34

The core model has 12.7% lower holdout MAE. We preserve the main model chosen on July-August instead of switching after observing September-October. The reversal is evidence that model rankings vary across time. Both features and target formulation differ, so this comparison cannot isolate the benefit of removing features. The holdout has now been inspected for diagnostics; it must not be reused as fresh tuning data.

What rate-per-mile MAE optimizes

The core candidate learns posted_rate / distance and converts predictions back to total dollars before evaluation. Because distance is positive, its unweighted training objective is proportional to abs(total_rate_error) / distance. A dollar of error on a shorter trip receives more weight than a dollar on a longer trip: this is a modeling choice, not just units.

This formulation was tried to represent distance-normalized pricing. On July-August total-dollar MAE, it scored $159.47 versus $163.01 for the core total-rate candidate. Weighting each per-mile training loss by distance would align the objective with total-dollar MAE, up to a constant. Section 8 reports this additional experiment.

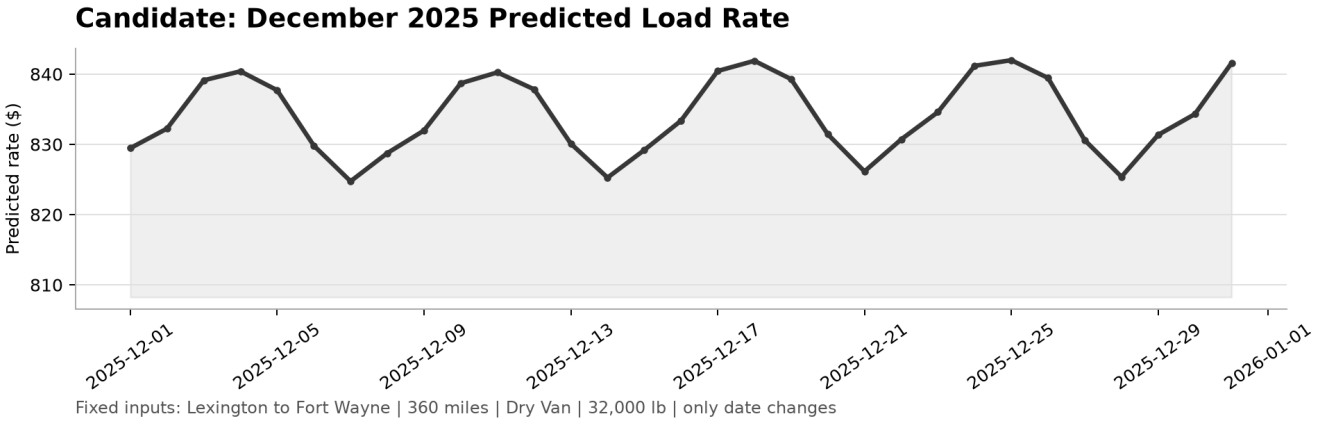
Unseen-city coverage and stress testing

The final set has 1,447 loads involving unseen cities (12.1%); the chronological holdout has none. Section 8 now tests deliberately withheld cities in July-August, excluding both pickup and delivery occurrences from fitting. This measures a simulated geographic shift, not accuracy for the eight actual unseen final cities.

Additional development compares feature sets and target formulations on earlier rolling chronological windows. These experiments were designed after the original holdout was inspected and are not a new untouched evaluation. Bias correction remains future work; the submitted models and predictions remain fixed.

6. Fixed December prediction chart

The image below is a monochrome rendering of candidate_december.png created by the supplied score.py from the completed December CSV. Only the date changes; all five other scenario inputs stay fixed.



The chart uses the separately evaluated six-input model. Unknown market and quote values are not filled with invented forecasts. The line describes model point predictions, not observed rates or an independently verified December seasonal pattern.

7. Reproducibility

Run solution.py to train and export models, predictions and metrics. Run the original score.py with the final and December CSVs to validate all 12,000 IDs, positive finite rates, the exact column schemas, and all 31 fixed December scenarios. Run test_solution.py for label/ID isolation, invalid-weight behavior and saved-model prediction parity. Run diagnostics.py for residual/geographic diagnostics, then build_report.py for this report.

Repository: github.com/fuad-mahee/spotter-freight-rate-assessment. The root README links the code, dependencies, saved models, predictions and run instructions.

8. Additional development experiments

These experiments were added after the original September-October results were known. They use labels only through August and do not reselect the submission models or calibrate predictions. Their role is to test sensitivity and guide future work; no new unbiased test score is claimed. The experiment protocol and row-level predictions are saved.

Controlled feature and loss comparison

Cross two feature sets (no-quote main inputs; six core inputs) with three objectives (total-dollar MAE; unweighted rate-per-mile MAE; distance-weighted rate-per-mile MAE). Tree settings and cleaning remain fixed. Fit all prior months, then evaluate Mar-Apr, May-Jun and Jul-Aug. These two-month horizons match the final task. Rank by equal-weight mean fold dollar MAE; retain all extreme targets. Windows differ in training size and regime.

Features / objective	Mean MAE ($)	Worst MAE ($)
Main inputs / RPM	128.03	140.55
Main inputs / weighted RPM	128.34	139.51
Main inputs / dollars	129.13	140.48
Core inputs / weighted RPM	141.94	163.74
Core inputs / RPM	143.14	159.47
Core inputs / dollars	143.47	163.01

Bias also changes sign. Original main specification mean errors: Mar-Apr: -$78.05, May-Jun: -$106.14, Jul-Aug: +$54.67. A single additive correction learned in one period may therefore fail in another. No bias offset is applied.

Geographic stress test

Choose six cities from January-June city names using each of three fixed seeds (17, 29, 43), without using targets or errors. Remove every training load touching those cities; evaluate July-August loads touching them. Compare the original main specification with the temporal development leader. Repetitions can overlap and are not independent samples.

Candidate	Mean stress-test MAE ($)
no_quote_direct	139.88
no_quote_per_mile	145.93

Interpretation: this is evidence about robustness, not proof of November-December accuracy. The original submission choice is preserved because no new untouched future labels are available. All per-fold results, selected cities, residual summaries and row-level predictions are available under output/robustness. Run robustness.py before rebuilding this report.